

Calorimetry 1

Essential Pre-Uni Chemistry F1.1

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level C1

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

Part A

(a)

Calculate the heat capacity of an object with mass 1.80 kg and specific heat capacity $0.32 \text{ J g}^{-1} \text{ K}^{-1}$.

Part B

(b)

Calculate the heat capacity of a calorimeter if its temperature is raised 2.5 K by 35 kJ of heat.

Part C

(c)

Calculate the expected increase in temperature when 2.4 kJ of heat is transferred to a calorimeter of heat capacity 720 J K^{-1} .

Part D

(d)

Calculate the heat required to raise the temperature of a calorimeter of heat capacity 1.6 kJ K^{-1} by 3.8°C .

Part E

(e)

Calculate the specific heat capacity of a calorimeter if it has a mass of 375 g and its temperature is raised 4.2 K by 2160 J of heat.

Part F

(f)

Calculate the heat required to raise the temperature of 3.14 kg of water by 12.2 K .

Part G

(g)

Calculate the mass of water whose temperature is raised through 16.0 K by 6.7 kJ of heat.

Part H

(h)

A calorimeter consists of 140 g of aluminium and 300 g of water. 6100 J of heat raises its temperature by 4.42 K. Calculate to three significant figures:

its heat capacity;

the specific heat capacity of aluminium.

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Calorimetry 3

Essential Pre-Uni Chemistry F1.3

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

The enthalpy change of combustion of naphthalene is $-5156 \text{ kJ mol}^{-1}$. Its molar mass is 128.2 g mol^{-1} . Calculate the temperature change expected when 1.00 mmol is burnt in excess oxygen in a calorimeter containing 4.0 kg of water.

Question deck:

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Calorimetry 5

Essential Pre-Uni Chemistry F1.5

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

Complete combustion of 0.020 mol of ethane, with a standard enthalpy of combustion of $-1410.8 \text{ kJ mol}^{-1}$ raises the temperature of the water in an insulated calorimeter from 17.4°C to 22.4°C . Calculate the mass of the water in the calorimeter.

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Calorimetry 7

Essential Pre-Uni Chemistry F1.7

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P2

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

25.0 cm^3 of sulfuric acid at 1.00 mol dm^{-3} and 19.10°C is placed in an insulated polystyrene cup. When 25.0 cm^3 of sodium hydroxide at 2.00 mol dm^{-3} and 19.10°C is added, the temperature rises to 32.45°C .

Assuming that no heat is lost, that the specific heat capacity of water may be used, and that the solutions have a density of 1.00 g cm^{-3} at 19.10°C , find the enthalpy change of the reaction per mole of water produced by neutralisation.

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Calorimetry 8

Essential Pre-Uni Chemistry F1.8

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P2

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$

30.0 cm^3 of ethanoic acid at 1.60 mol dm^{-3} and 18.65°C is placed in an insulated polystyrene cup. When 40.0 cm^3 of sodium hydroxide at 1.00 mol dm^{-3} and 18.65°C is added, the temperature rises to 25.80°C .

Assuming that no heat is lost, that the specific heat capacity of water may be used, and that the solutions have a density of 1.00 g cm^{-3} at 18.65°C , find the enthalpy change of the reaction per mole of water produced by neutralisation.

Question deck:

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Calorimetry 9

Essential Pre-Uni Chemistry F1.9

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

When 5.0 g of ammonium nitrate dissolves in 100 g of water, the temperature of the water drops from 18°C to 14°C . Calculate the enthalpy of solution of ammonium nitrate in kJ mol^{-1} using the following scheme.

Part A

Formula

Write down the formula of ammonium nitrate.

Part B

Relative formula mass

Calculate the relative formula mass of ammonium nitrate. Give your answer to 3 significant figures.

Part C

Number of moles

Calculate the number of moles of ammonium nitrate in 5.0 g.

Part D

Heat loss of water

Calculate the heat lost from the 100 g of water. Give your answer to 2 significant figures.

Part E

Molar heat loss of ammonium nitrate

Calculate the heat lost per mole of ammonium nitrate.

Part F

Enthalpy of ammonium nitrate

Give the enthalpy of solution of ammonium nitrate.

Question deck:

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Calorimetry 10

Essential Pre-Uni Chemistry F1.10

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

The enthalpies of combustion of three fuels are shown below:

Fuel	$\Delta H_c / \text{kJ mol}^{-1}$
CH_4	−890.3
C_3H_8	−2219.2
C_4H_{10}	−2876.5

Part A

(a)

Which gives out most heat per gram?

Part B

(b)

Which gives out most heat per mole?

Part C

(c)

Which gives out most heat per cubic foot?

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STEM SMART Chemistry Week 12

Average Bond Enthalpy

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

The average bond enthalpy of the C–F bond in tetrafluoromethane is given by the standard enthalpy change for one of the following reactions. Which one is it?

- ☐ $\text{CF}_4(\text{s}) \longrightarrow \text{CF}_4(\text{g})$
- ☐ $\frac{1}{4} \text{CF}_4(\text{g}) \longrightarrow \frac{1}{4} \text{C}(\text{g}) + \text{F}(\text{g})$
- ☐ $\text{CF}_4(\text{g}) \longrightarrow \text{CF}_3(\text{g}) + \text{F}(\text{g})$
- ☐ $\text{CF}_4(\text{g}) \longrightarrow \text{CF}_3^+(\text{g}) + \text{F}^-(\text{g})$
- ☐ $2 \text{F}_2(\text{g}) + \text{C}(\text{s}) \longrightarrow \text{CF}_4(\text{g})$

Adapted with permission from UCLES, OCSEB A Level Chemistry, June 1995, Paper 1, Question 10

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Bond Enthalpies 3

Essential Pre-Uni Chemistry F2.3

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** GCSE C3, A Level P1

Use some of the following bond enthalpies in kcal mol^{-1} to calculate the enthalpy changes for the reactions (in the gas phase):

C=C	146	C $\equiv$ O	258
O=O	119	H–O	111
C–H	99	H–H	104
C=O	178		

Part A

(a)

$\text{C}_2\text{H}_4 + \text{O}_2 \longrightarrow 2 \text{CH}_2\text{O}$. Give your answer to 2 significant figures.

Part B

(b)

$\text{CO} + \text{H}_2\text{O} \longrightarrow \text{CO}_2 + \text{H}_2$. Give your answer to 2 significant figures.

Part C

(c)

$\text{CH}_2\text{O} \longrightarrow \text{CO} + \text{H}_2$. Give your answer to 2 significant figures.

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Bond Enthalpies 4

Essential Pre-Uni Chemistry F2.4

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

Given that the bond energy of H–H is 4.53 eV, D–D is 4.59 eV, and the energy change on reaction $\text{H}_2 + \text{D}_2 \longrightarrow 2 \text{HD}$ is +0.02 eV, find the bond energy of H–D. Give your answer to 3 significant figures.

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